

LegacyLift AI

The Operating System for Autonomous Legacy System Modernization

One-liner: The AI platform that understands, documents, and transforms enterprise legacy systems—turning 50-year-old COBOL mainframes into modern cloud-native applications without breaking the business.

The \$2 Trillion Time Bomb

Every day, \$3 trillion in transactions flow through systems written in COBOL. Banks, insurers, governments, and Fortune 500 companies run on code older than their youngest employees. The problem? The developers who built these systems are retiring at 10,000 per day, and no one is learning COBOL anymore.

The crisis is now: - 95% of ATM transactions touch COBOL code - 80% of in-person transactions use COBOL - The average mainframe developer is 58 years old - Estimated \$800B in technical debt locked in legacy systems - 75% of Fortune 500 modernization projects fail

The old approach is broken: - Manual code archaeology takes years - “Lift and shift” creates cloud-hosted legacy nightmares - Complete rewrites cost hundreds of millions and often fail - Consulting firms charge \$500/hr and still can’t deliver

LegacyLift AI is the answer: An autonomous platform that understands legacy systems at the architectural, business logic, and code level—then transforms them into modern applications with zero business disruption.

How It Works

1. Deep System Understanding

- AI agents that can read and comprehend COBOL, PL/I, Fortran, AS/400 RPG, Natural/ADABAS, and 200+ legacy languages
- Automatic discovery of business rules buried in 40-year-old code
- Dynamic documentation generation—turning tribal knowledge into living specs
- Data lineage mapping across decades of accumulated systems

2. Intelligent Decomposition

- AI identifies bounded contexts and microservice boundaries
- Automatic dependency mapping and impact analysis
- Risk scoring for every component and transformation
- Prioritization engine that sequences modernization for maximum value, minimum risk

3. Autonomous Transformation

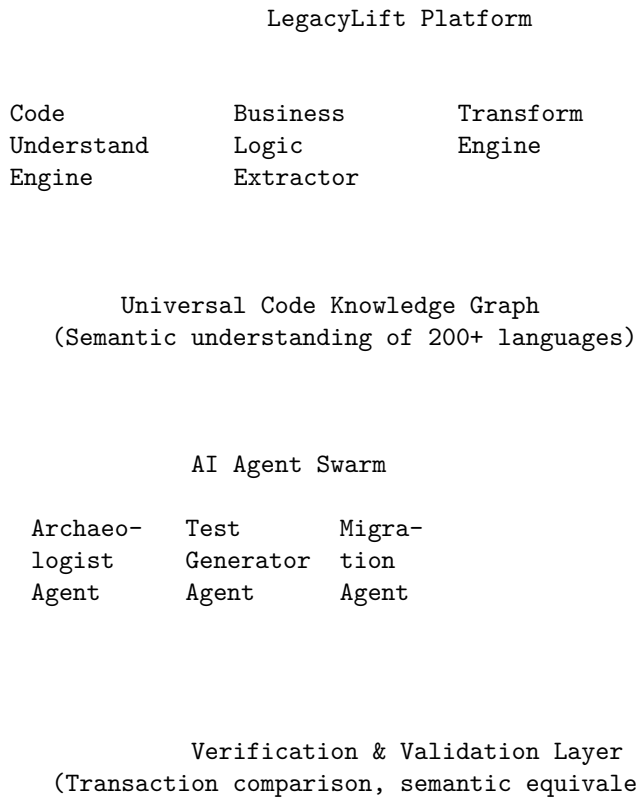
- Code-to-code translation with semantic preservation
- AI-generated test suites that validate business logic preservation
- Parallel running capability—old and new systems compared transaction-by-transaction
- Gradual cutover with automatic rollback

4. Continuous Modernization

- Post-migration optimization and refactoring
- Technical debt prevention through continuous analysis
- AI pair programming for ongoing maintenance

- Knowledge preservation as legacy experts retire

The Technology Stack



Core Innovations

Universal Code Understanding Model (UCUM) - Foundation model trained on 50+ years of enterprise code - Understands not just syntax but business intent - Cross-language semantic mapping - Trained on successful modernization patterns from thousands of projects

Autonomous Archaeologist Agents - AI agents that explore codebases like archaeologists explore dig sites - Interview remaining experts and extract tacit knowledge - Reconstruct lost documentation from code behavior - Build institutional memory before it's lost forever

Semantic Equivalence Engine - Mathematical proofs that transformed code preserves business logic - Property-based testing at scale - Behavioral fingerprinting and comparison - Automated regulatory compliance verification

Continuous Migration Orchestrator - Manages gradual cutovers across millions of transactions - Real-time performance and accuracy monitoring - Automatic anomaly detection and rollback - Zero-downtime transformation

Market Opportunity

The Numbers Are Staggering

Segment	Market Size	Growth
Mainframe Modernization	\$24B (2026)	14% CAGR
Application Modernization	\$51B (2026)	16% CAGR
Legacy System Support	\$38B (2026)	8% CAGR
Technical Debt Services	\$15B (2026)	22% CAGR
Total Addressable Market	\$128B	14% CAGR

Customer Segments

Tier 1: Financial Services - Banks with 50+ year old core systems - Insurance companies running claims on mainframes - Payment processors built on COBOL - Average deal size: \$5-50M annually

Tier 2: Government - IRS, Social Security, state DMVs - Defense systems on ancient infrastructure - Critical infrastructure SCADA systems - Average deal size: \$10-100M (multi-year)

Tier 3: Healthcare & Life Sciences - Hospital systems on AS/400 - Pharmaceutical R&D data on legacy platforms - Insurance claims processing - Average deal size: \$2-20M annually

Tier 4: Manufacturing & Logistics - ERP systems from the 1980s - Supply chain systems on proprietary platforms - Industrial control systems - Average deal size: \$1-10M annually

Business Model

Revenue Streams

1. Platform SaaS (40% of revenue) - Code understanding and documentation: \$50K-500K/month - Transformation engine access: \$100K-1M/month - Continuous modernization: \$25K-250K/month

2. Outcome-Based Pricing (35% of revenue) - Per-function modernized - Per-line of code transformed - Per-system decommissioned - Success bonuses tied to go-live

3. Managed Modernization Services (20% of revenue) - End-to-end transformation delivery - AI + human expert teams - Fixed-price or T&M engagements

4. Training & Certification (5% of revenue) - Enterprise team enablement - LegacyLift certification program - Knowledge transfer workshops

Unit Economics

Metric	Year 1	Year 3	Year 5
Average Contract Value	\$2M	\$5M	\$8M
Gross Margin	65%	75%	82%
Net Revenue Retention	140%	160%	175%
CAC Payback	18 mo	12 mo	8 mo
LTV:CAC	4x	8x	12x

Competitive Landscape

Current Players & Their Limitations

Competitor	Approach	Limitation
Accenture/IBM	Army of consultants	\$500/hr, multi-year timelines, 60% failure rate
Micro Focus AWS Migration Hub	Legacy runtime extension Cloud lift-and-shift tools	Kicks the can, doesn't modernize Creates cloud-hosted legacy nightmares
BluAge (AWS) Modern Systems	Rule-based code translation COBOL-to-Java translation	Limited language support, brittle Produces unreadable code, no semantic understanding

Why LegacyLift Wins

10x Faster: AI understands code in hours, not months **10x Cheaper:** Autonomous agents vs. \$500/hr consultants **10x Safer:** Semantic equivalence proofs, not “trust us” **10x Smarter:** Learns from every project, gets better continuously

Go-to-Market Strategy

Phase 1: Lighthouse Customers (Months 1-12)

- Partner with 5 Fortune 500 companies for joint development
- Target companies with urgent modernization deadlines
- Build case studies and ROI proof points
- Focus: Financial services (highest urgency, deepest pockets)

Phase 2: Vertical Expansion (Months 12-24)

- Launch government-specific offering (FedRAMP)
- Healthcare compliance package (HIPAA)
- Manufacturing/industrial package
- Build partner ecosystem with SIs

Phase 3: Platform Ecosystem (Months 24-36)

- Open marketplace for transformation patterns
- Third-party agent development
- Consulting partner certification program
- International expansion (UK/EU legacy crisis)

Phase 4: Autonomous Operations (Months 36-48)

- Fully autonomous modernization for standard patterns
 - Self-service for mid-market companies
 - Acquisition of legacy talent agencies
 - Category dominance
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The Team We Need

Founding Team

CEO - Enterprise software veteran with CIO relationships **CTO** - Compiler/PL researcher with production experience **Chief Scientist** - LLM expert, code understanding focus **Chief Revenue Officer** - Enterprise sales leader (IBM/Accenture) **Head of Delivery** - Modernization project veteran

Key Early Hires

- COBOL/mainframe archaeologists (yes, they're rare and expensive)
- Compiler engineers
- ML infrastructure engineers
- Enterprise solution architects
- Customer success managers

Advisory Board

- Retired CIOs from major banks
 - Mainframe legends (the people who built these systems)
 - AI safety researchers (for transformation verification)
 - Former regulators (OCC, Fed, SEC)
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Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Customers	5	20	60	150	350
ARR	\$10M	\$50M	\$180M	\$500M	\$1.2B
Revenue	\$8M	\$45M	\$165M	\$450M	\$1.1B
Gross Profit	\$5M	\$32M	\$124M	\$360M	\$902M
Operating Margin	-80%	-40%	5%	25%	35%
Employees	50	150	400	800	1,500

Funding Strategy

Seed Round: \$8M

- Build core platform and Universal Code Understanding Model
- 5 design partners
- Prove technology on real mainframes

Series A: \$40M

- Scale engineering and go-to-market
- First 20 enterprise customers
- FedRAMP certification

Series B: \$120M

- International expansion
- Partner ecosystem build-out
- Acquisition of legacy talent pools

Series C: \$300M

- Platform ecosystem expansion
- Market consolidation through M&A
- Path to profitability

Target Exit: \$10-15B (Strategic or IPO)

- Strategic acquirers: Microsoft, Google, IBM, Accenture
- IPO: Enterprise software comps at 15-20x ARR

Risk Factors & Mitigation

Risk	Probability	Impact	Mitigation
AI accuracy insufficient	Medium	High	Extensive validation, human-in-loop, gradual rollout
Legacy experts unavailable	High	Medium	Aggressive early hiring, knowledge capture priority
Enterprise sales cycles	High	Medium	Outcome-based pricing, land and expand
Regulatory scrutiny	Medium	High	SOC 2, FedRAMP, proactive regulator engagement
Competition from hyperscalers	Medium	Medium	Speed to market, specialization moat, enterprise relationships

Why Now?

1. **The talent crisis is peaking.** 10,000 COBOL developers retire daily. The window to capture institutional knowledge is closing.
2. **AI finally understands code.** LLMs can now comprehend business logic, not just syntax. Semantic transformation is possible.
3. **Enterprises are desperate.** Boards are mandating modernization. Compliance deadlines are forcing action.
4. **Cloud economics are compelling.** Mainframe costs are exploding. The business case for transformation is undeniable.
5. **The competition is weak.** Consulting firms are expensive and fail often. Point tools are primitive. No one owns this category.

The Vision

Today: Legacy systems are ticking time bombs—expensive, fragile, and increasingly unmaintainable.

2030: LegacyLift has transformed \$1 trillion in enterprise systems, freeing companies to innovate instead of maintain.

2035: Legacy modernization is a solved problem. Every system gets continuous modernization, like automatic software updates.

We're not building a modernization tool. We're building the **time machine** that rescues 50 years of enterprise computing from the past and brings it into the future.

The Ask

We're raising a **\$8M Seed round** to build the Universal Code Understanding Model and prove LegacyLift with 5 Fortune 500 design partners.

What we offer: - Category-defining opportunity in a \$128B market - Team with deep enterprise and AI expertise - Clear technical moat and go-to-market strategy - Path to \$1B+ ARR within 5 years

The time is now. Every day we wait, more institutional knowledge walks out the door forever.

"The best time to modernize legacy systems was 20 years ago. The second best time is with LegacyLift AI."

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